S7CAS-PLEIADES-CARTE-MERE

Pleiades

01

Revision 1.00

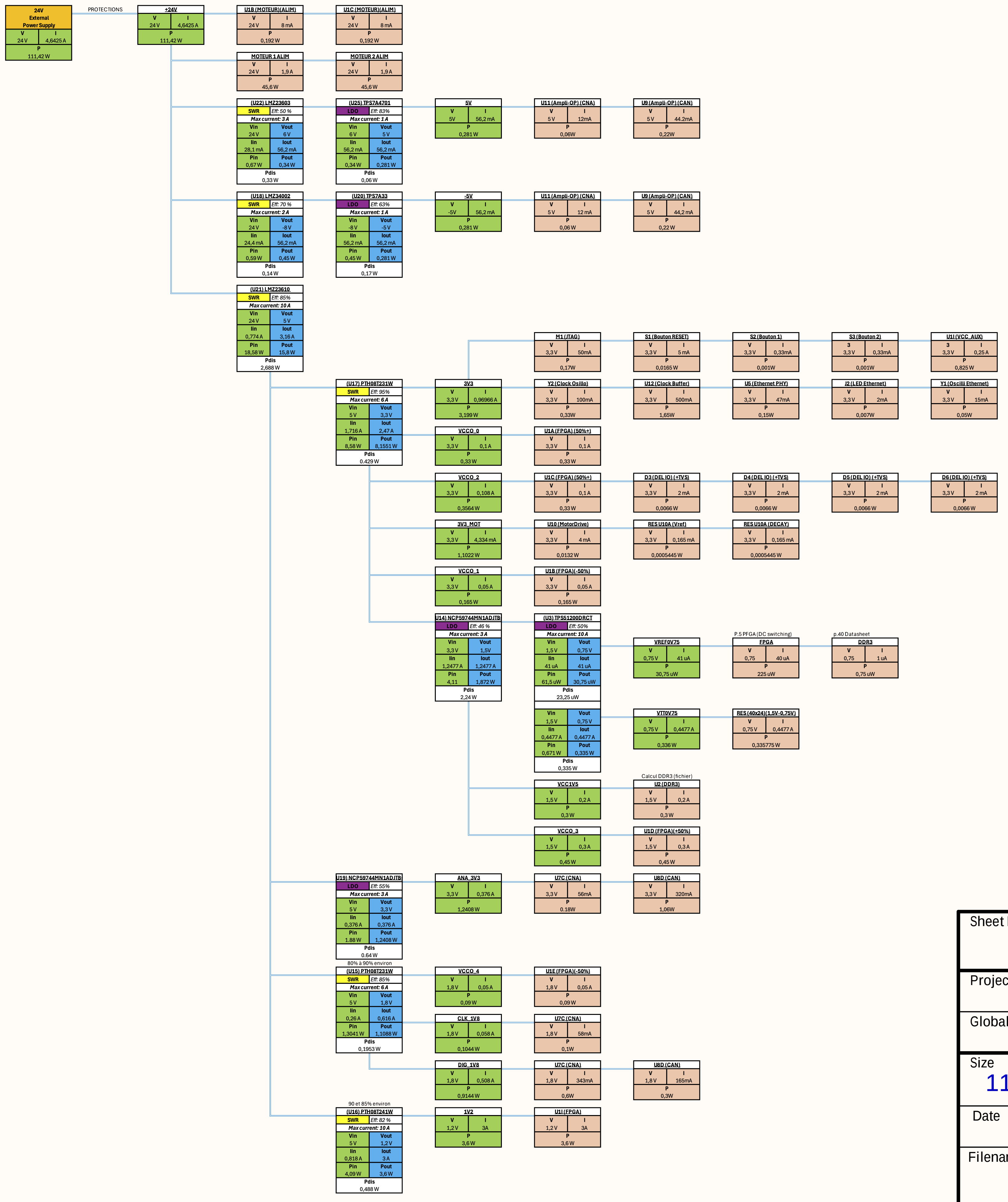
Date : 13/01/2026

Revision history	

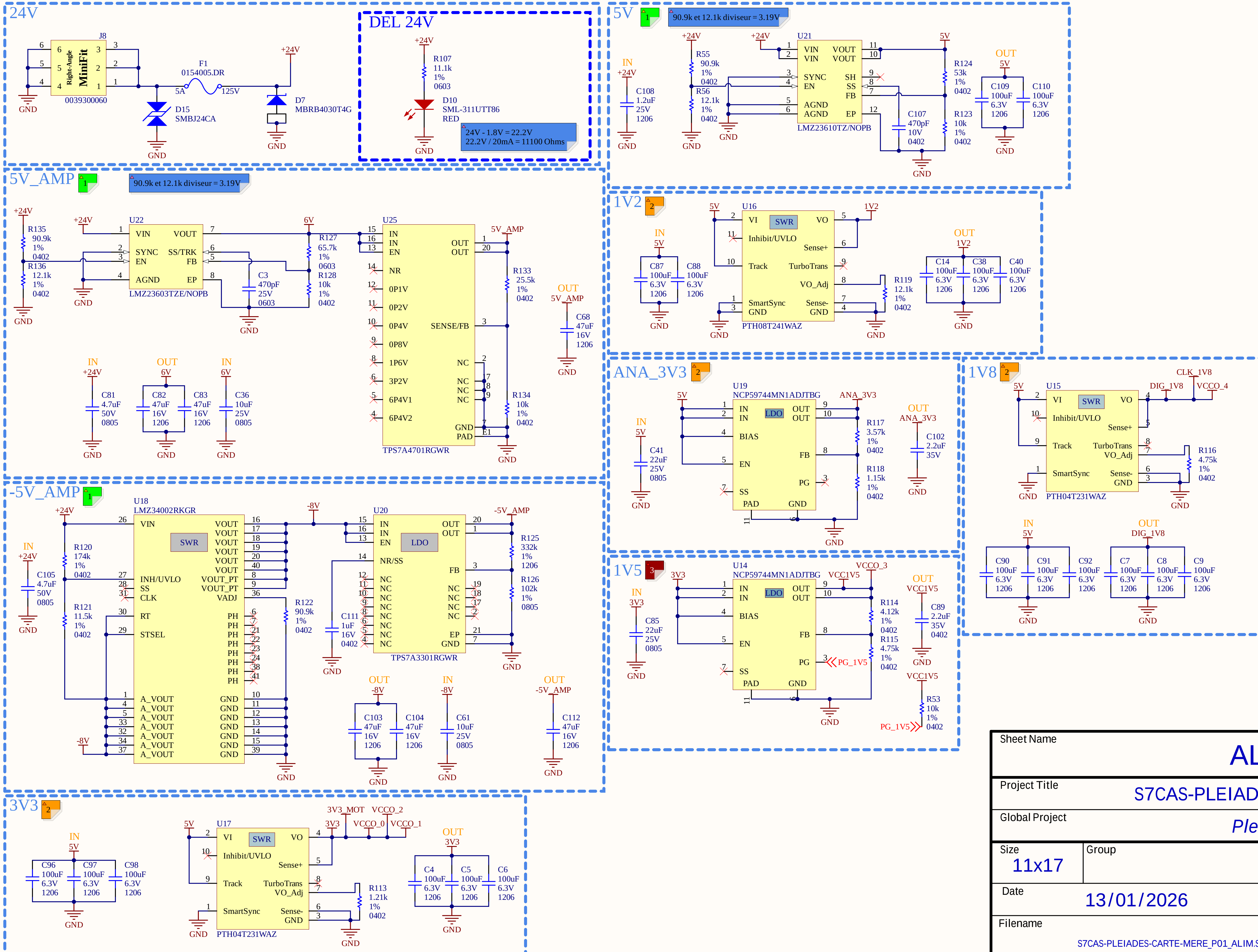
Package size conversion	
Metric	*Imperial*
1005	0402
1608	0603
2012	0805
3216	1206
3225	1210
6432	2512

FID4 FID 1MM
FID5 FID 1MM
FID6 FID 1MM
FID1 FID 1MM
FID2 FID 1MM
FID3 FID 1MM

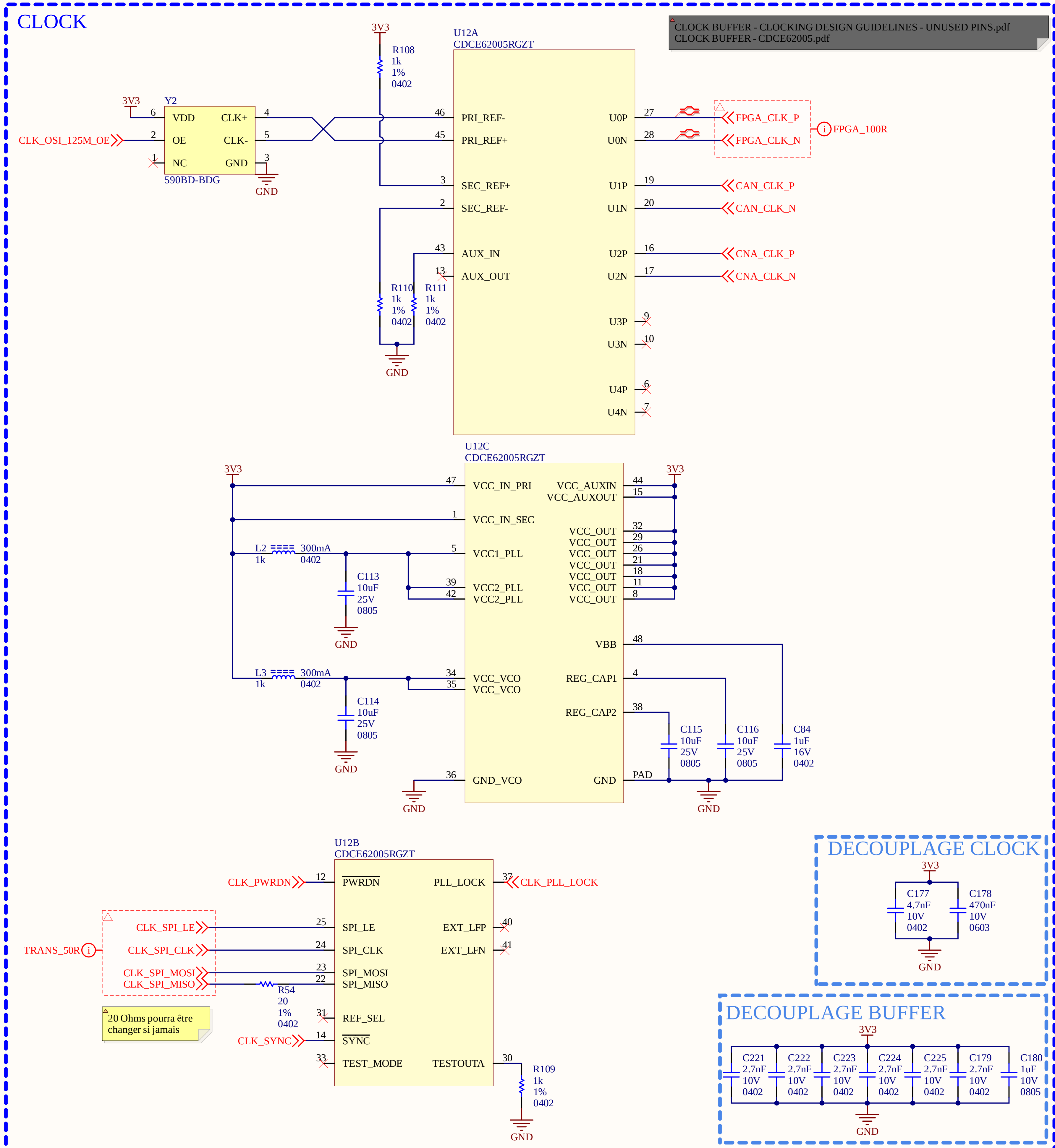
Project Title: S7CAS-PLEIADES-CARTE-MERE			
Global Project: *Pleiades*			
Size: 11x17	Group: 01		Revision: 1.00
Date: 13/01/2026		Sheet: 1 of 17	
Filename: S7CAS-PLEIADES-CARTE-MERE_P01_TITLE.SchDoc		Designers: Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard	

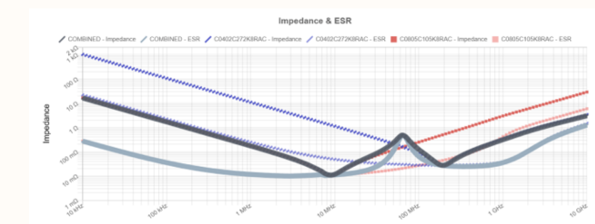


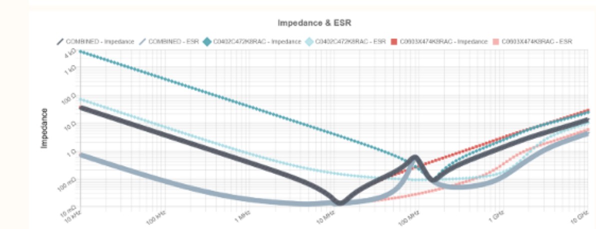
Sheet Name	POWER TREE		
Project Title	S7CAS-PLEIADES-CARTE-MERE		
Global Project	*Pleiades*		
Size: 11x17	Group: 01		Revision: 1.00
Date: 13/01/2026		Sheet: 2 of 17	
Filename: S7CAS-PLEIADES-CARTE-MERE_P01_POWER.SchDoc		Designers: Mathieu Desautels (DESM1210), Mathys Plante (PLAM1001), Jonathan Bouchard	



Sheet Name	ALIM
Project Title	S7CAS-PLEIADES-CARTE-MERE
Global Project	*Pleiades*
Size	11x17
Group	01
Revision	1.00
Date	13/01/2026
Sheet	3 of 17
Filename	S7CAS-PLEIADES-CARTE-MERE_P01_ALIM.SchDoc
Designers	Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard

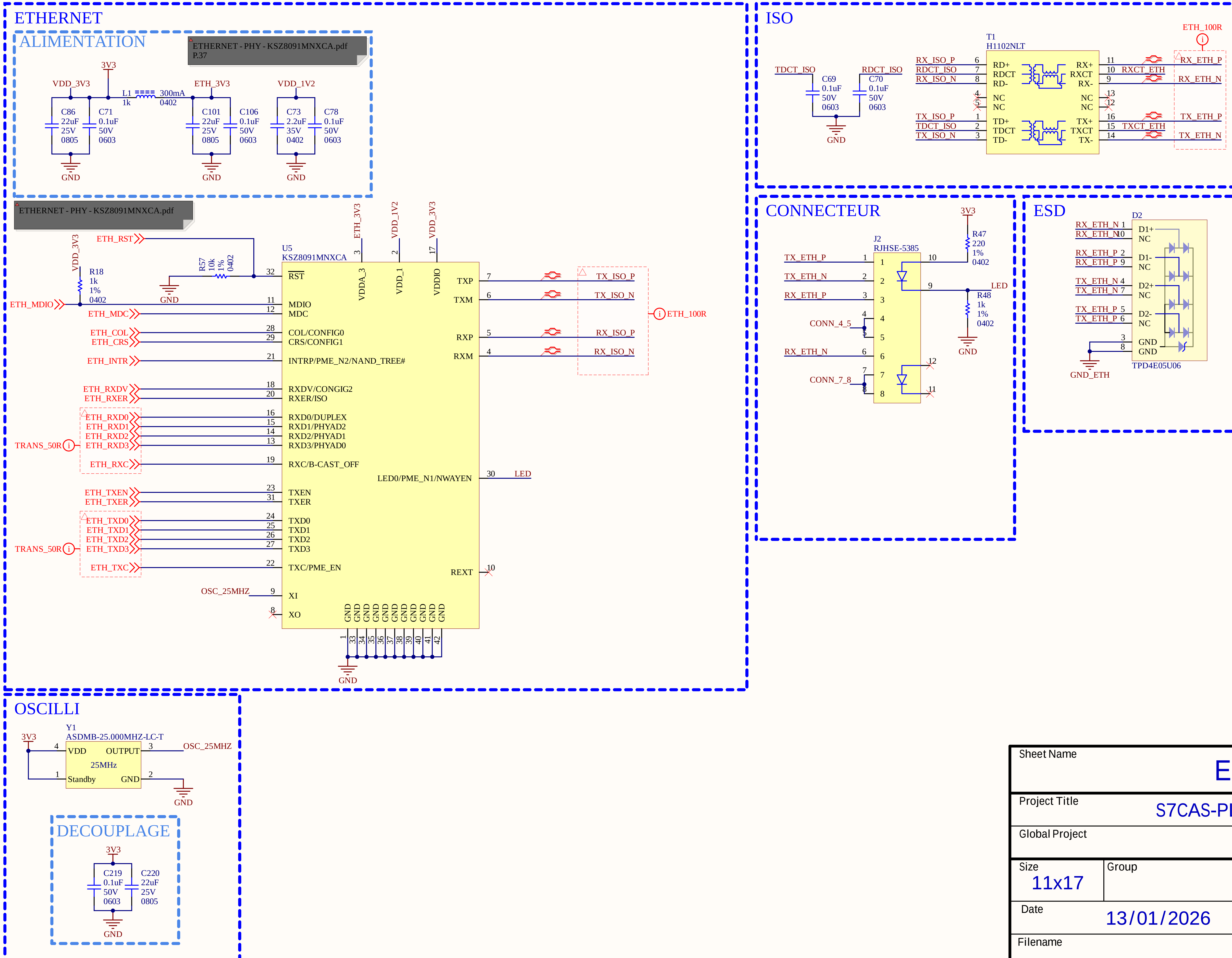






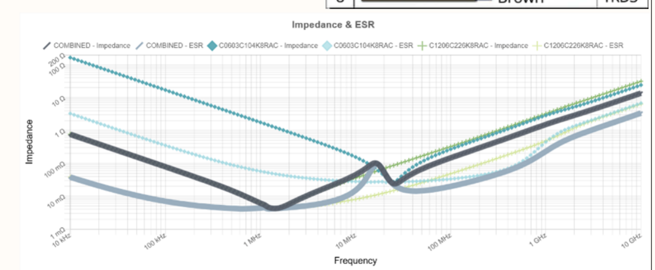
-Quand tu es en LVDS faut que branche le SEC à une résistance (CLOCK BUFFER - CLOCKING DESIGN GUIDELINES - UNUSED PINS.pdf)
-CAN et CNA pas fait pour la pair différentiel puisqu'elle est directement dans le CAN et CNA

Sheet Name	CLOCK
Project Title	S7CAS-PLEIADES-CARTE-MERE
Global Project	*Pleiades*
Size	11x17
Group	01
Revision	1.00
Date	13/01/2026
Sheet	4 of 17
Filename	S7CAS-PLEIADES-CARTE-MERE_P01_CLOCK.SchDoc
Designers	Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard



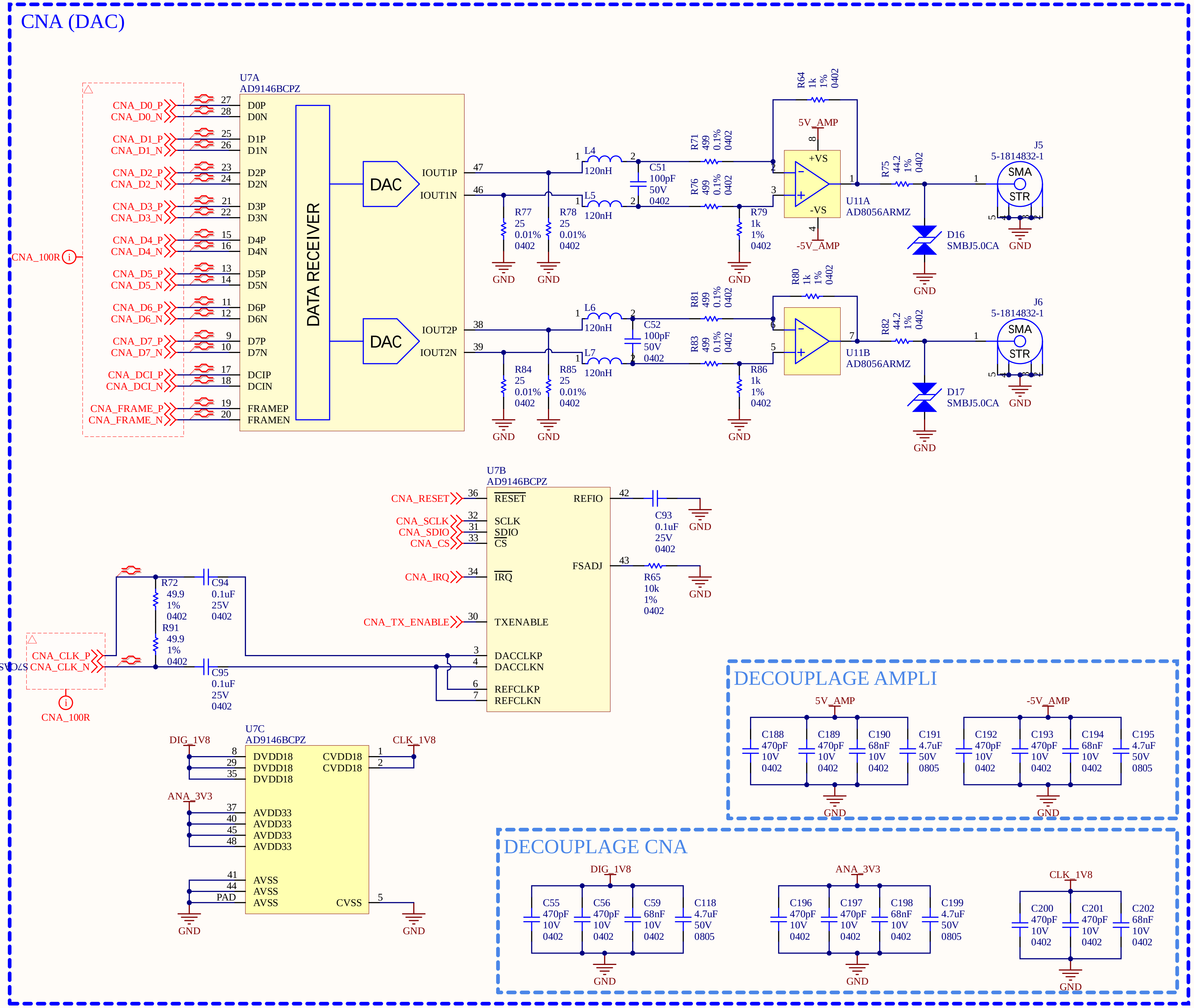
TIA 568A

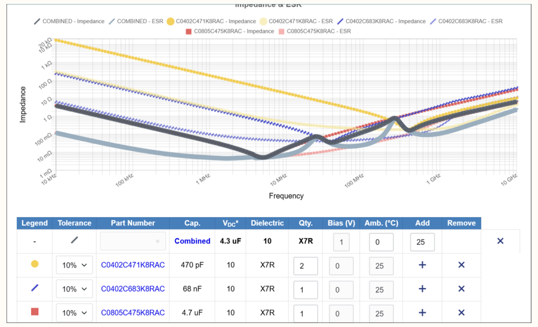
Pin #	Wire Color Legend	Signal
1	White/Green	TX+
2	Green	TX-
3	White/Orange	RX+
4	Blue	TRD2+
5	White/Blue	TRD2-
6	Orange	RX-
7	White/Brown	TRS3+
8	Brown	TRD3-



-Isolation du ETHERNET 1.5kV
-Ajout d'un condensateur de 1.5kV
-Ligne de transmission de 50R
-TX et RX voir le reférence design pourquoi c'est ligne de transmission sans différentiel

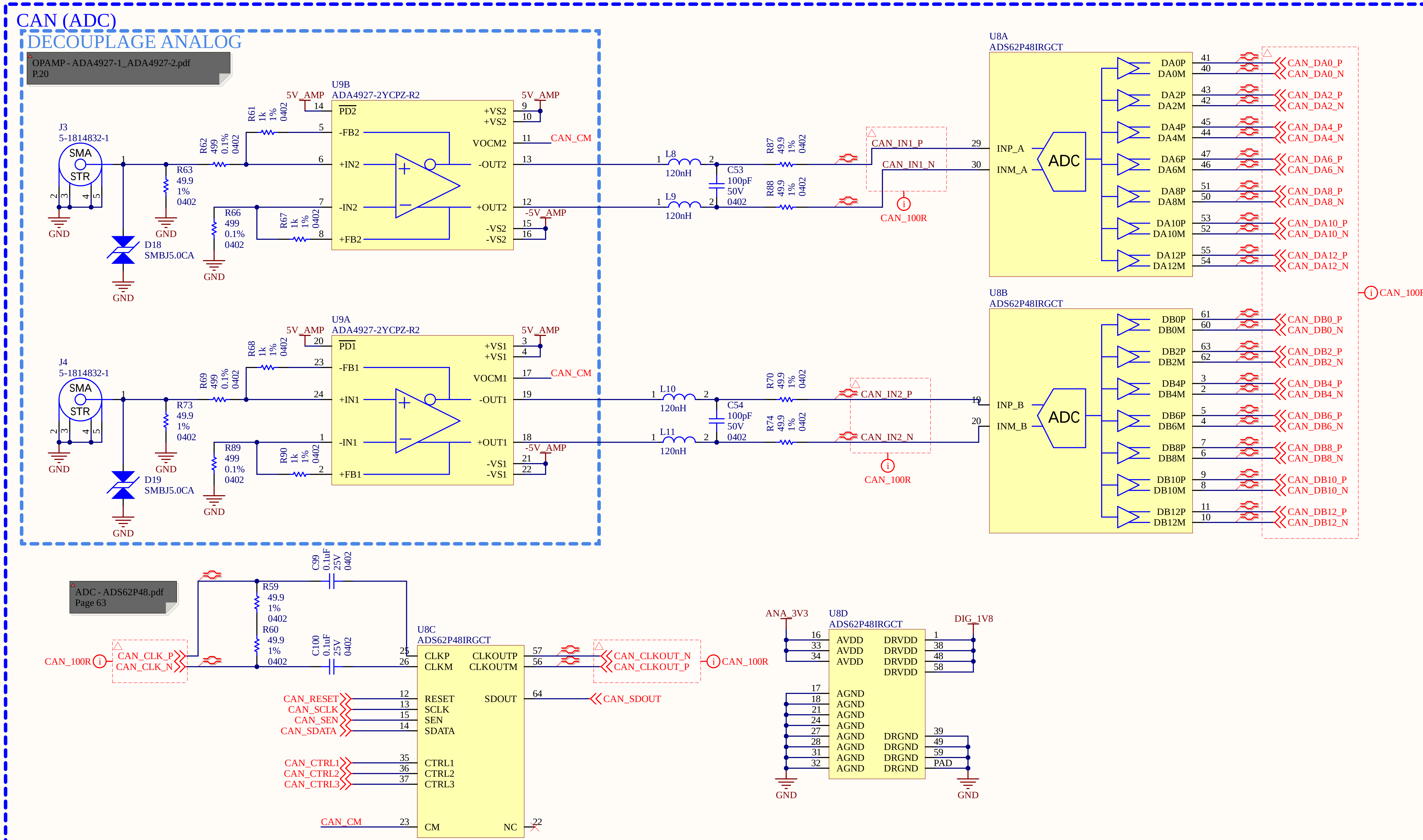
Sheet Name	ETHERNET
Project Title	S7CAS-PLEIADES-CARTE-MERE
Global Project	*Pleiades*
Size	11x17
Group	01
Revision	1.00
Date	13/01/2026
Sheet	5 of 17
Filename	S7CAS-PLEIADES-CARTE-MERE_P01_ETHERNET.SchDoc
Designers	Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard

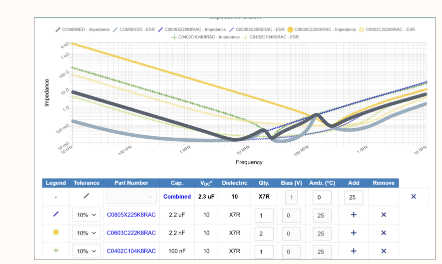




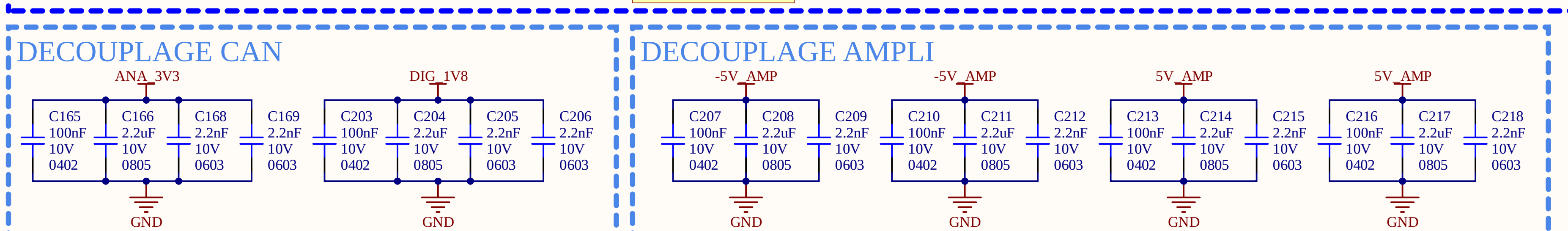
-200mbit/s en DDR LVDS
-Toute les données vont sur la BANK 1
-Toute le control sur bank 4
- Ampli-op aimerait un gain de 2 avec 1V à la sortie, il faut donc 0.5V en entree
-DAC sortie au maximum c'est 20mA, donc avec le 25ohms c'est 0.5V
-On reussi à avoir 1V à la sortie
-5ohms à la sortie à 40MHz donc 45ohms pour adapter
-Filtre passe bas de deuxième ordre (0 à 40MHz), point de coupure 45MHz = L et C
-Receiver Differential impedance 80 to 120ohms

Sheet Name	CNA
Project Title	S7CAS-PLEIADES-CARTE-MERE
Global Project	*Pleiades*
Size	11x17
Group	01
Revision	1.00
Date	13/01/2026
Sheet	6 of 17
Filename	S7CAS-PLEIADES-CARTE-MERE_P01_CNA.SchDoc
Designers	Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard

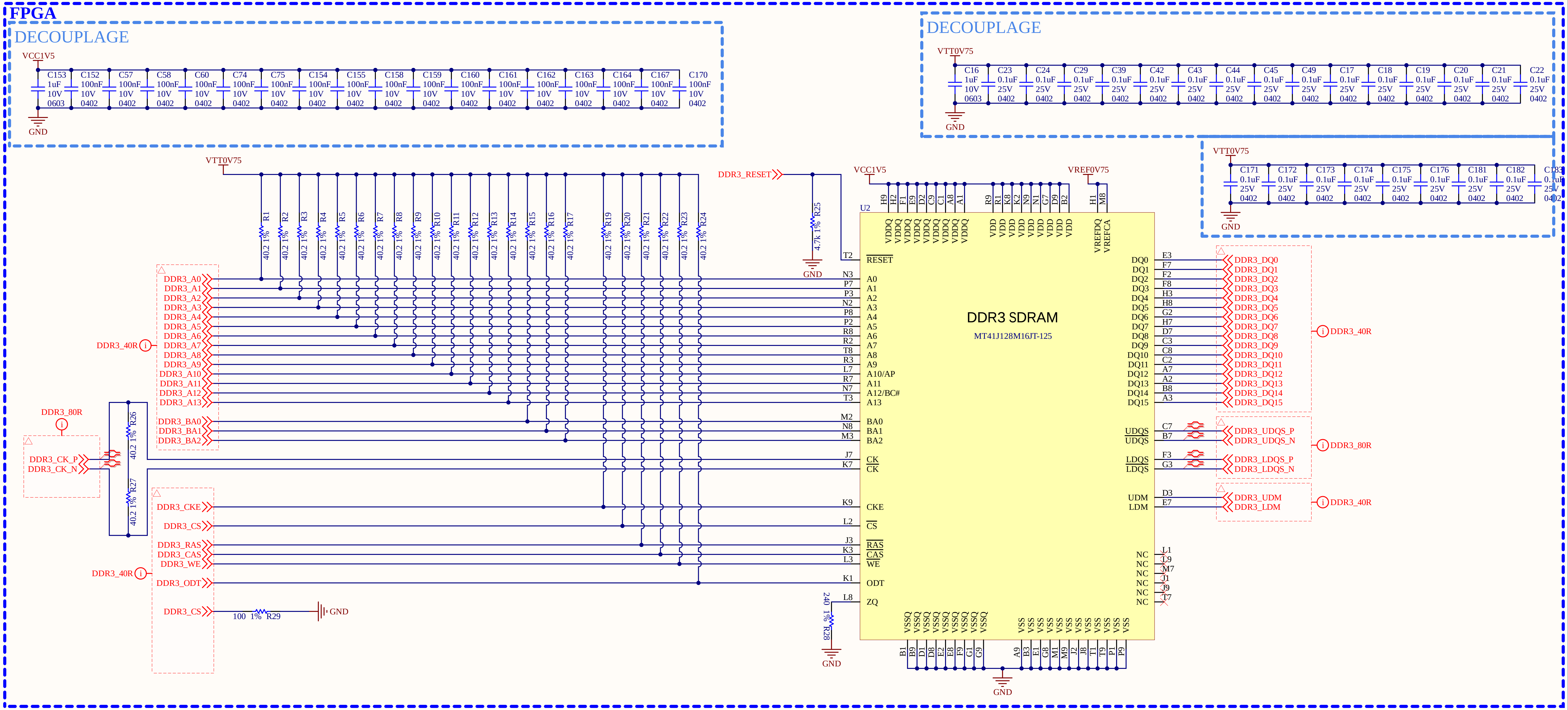


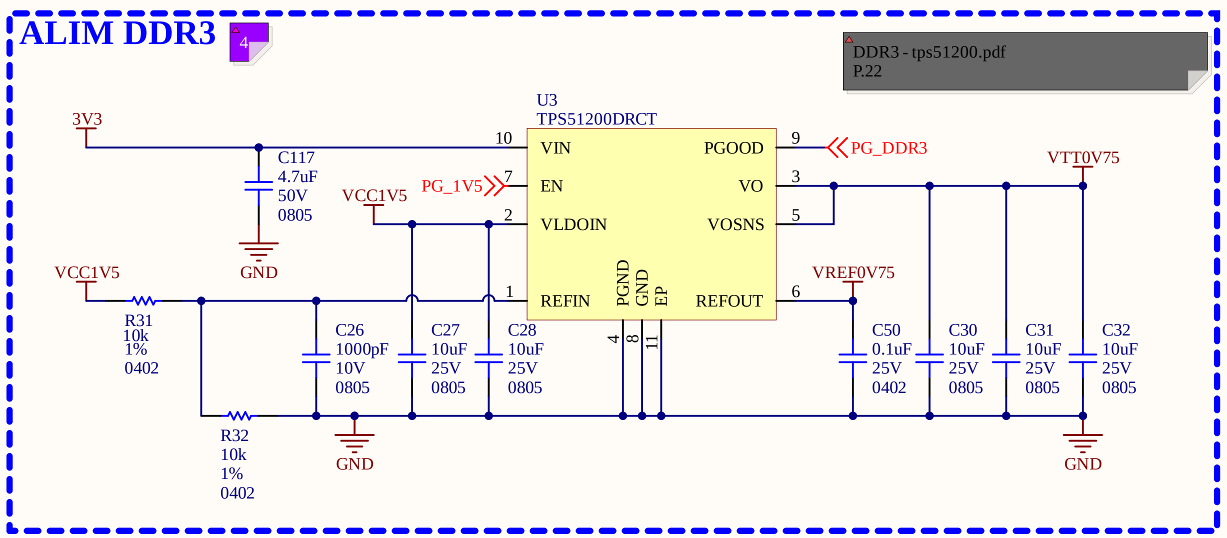


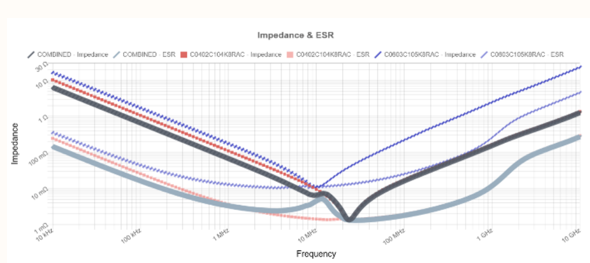
-Choix de OP-AMP pour isolé et pour passer d'analog a digital
-Toute les données vont sur la BANK 1
-Toute le control sur bank 4
- Nous avons pris uniquement AMPLI-OP puisque transfo bloque le DC et on a une composante DC



Sheet Name	CAN
Project Title	S7CAS-PLEIADES-CARTE-MERE
Global Project	*Pleiades*
Size	11x17
Group	01
Revision	1.00
Date	13/01/2026
Sheet	7 of 17
Filename	S7CAS-PLEIADES-CARTE-MERE_P01_CAN.SchDoc
Designers	Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard



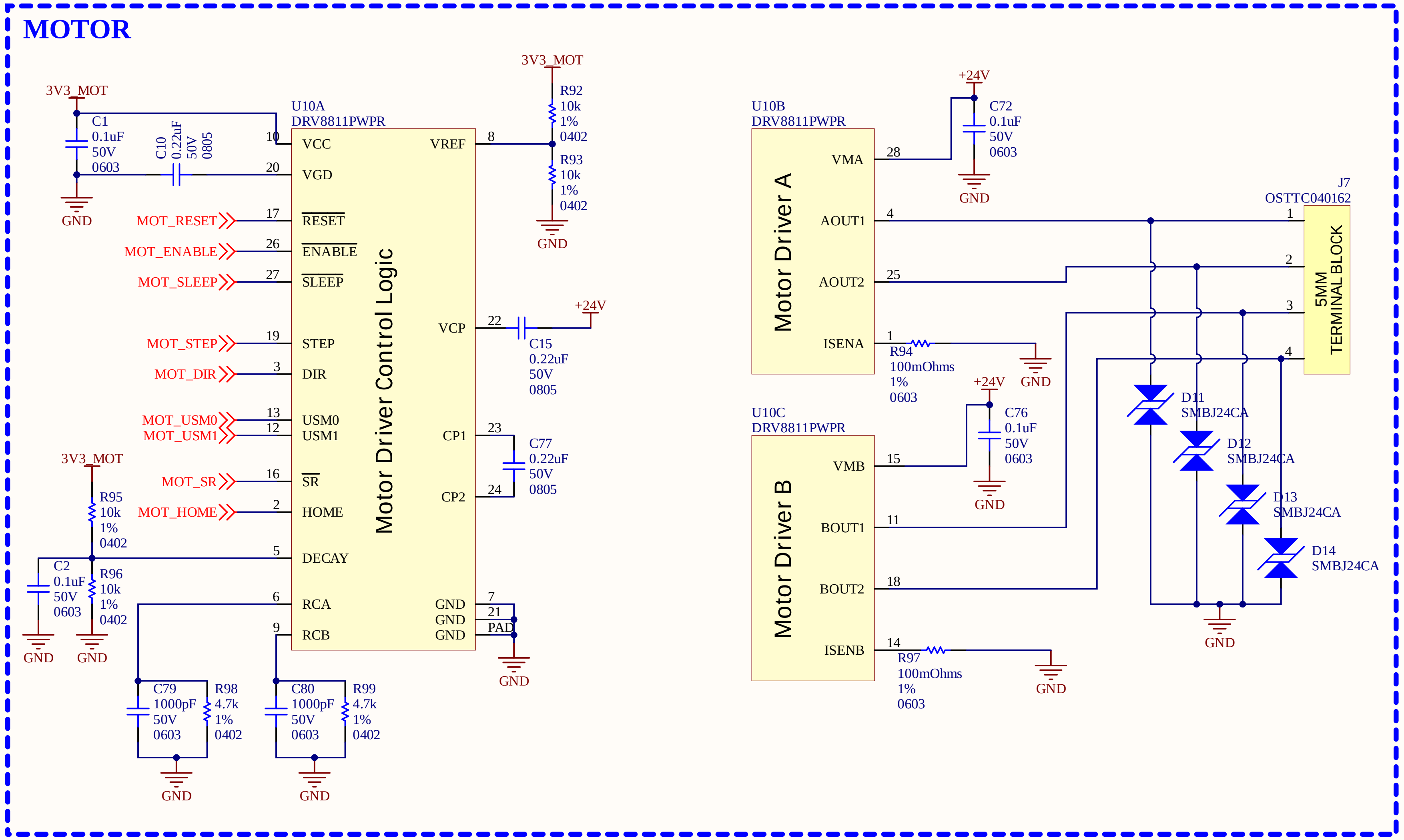


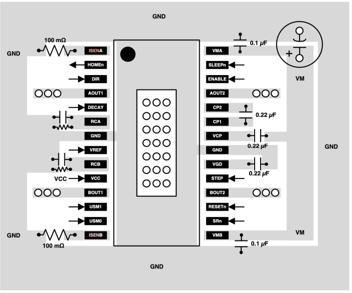


-Ligne de transmission de 40R
-Resistance paralèle sur la ligne de transmission de 40.2Ohms
-Clock en pair différentiel 80R
-Découplage il y 18 pins donc on peut en mettre 18 condensateurs
-Dans un cours on s'est fait dire que c'est un condensateur par résistance de VTT

Sheet Name	DDR3
Project Title	S7CAS-PLEIADES-CARTE-MERE
Global Project	Pleiades
Size	11x17
Group	01
Revision	1.00
Date	13/01/2026
Sheet	8 of 17
Filename	S7CAS-PLEIADES-CARTE-MERE_P01_DDR3.SchDoc
Designers	Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard

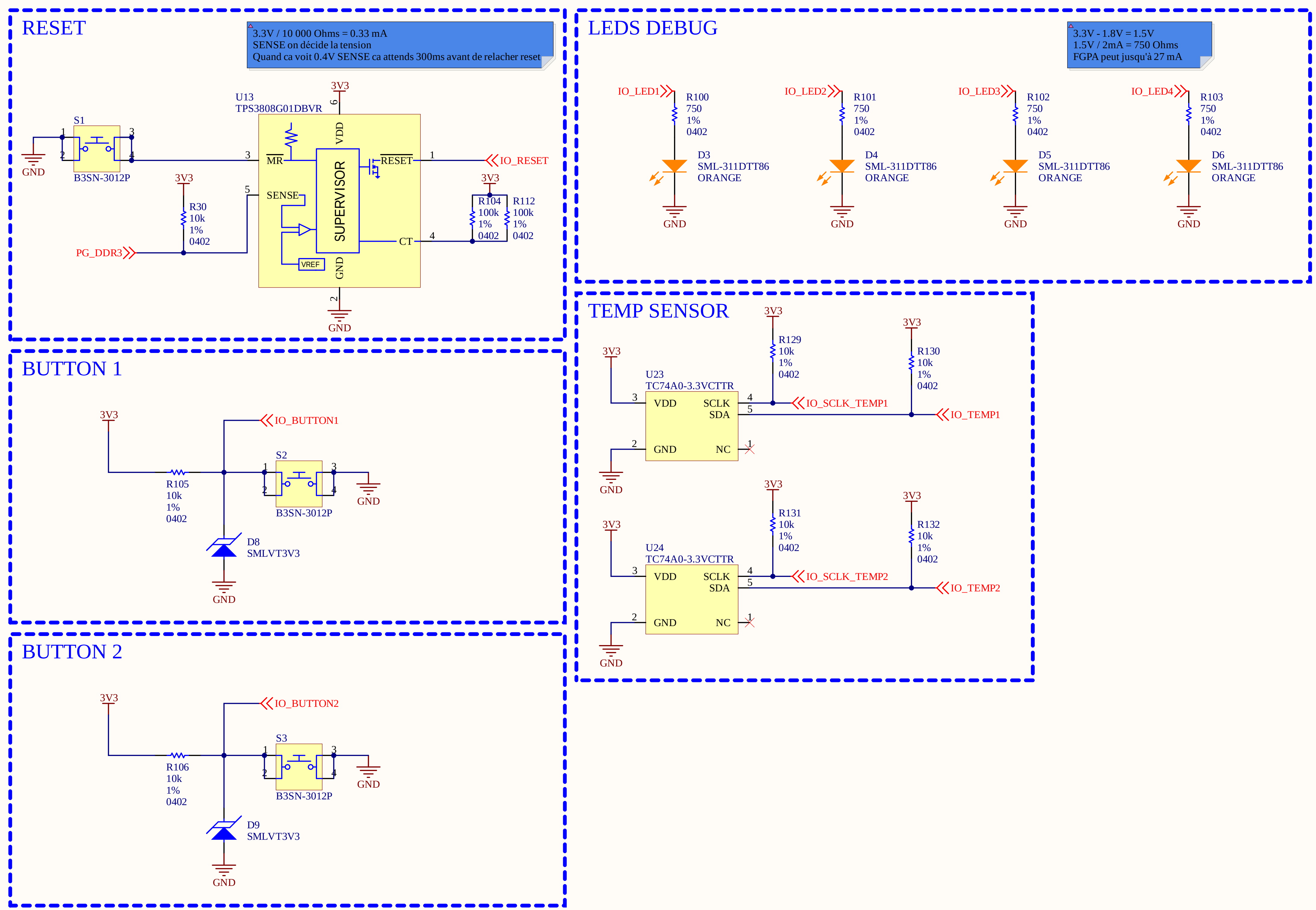
MOTOR



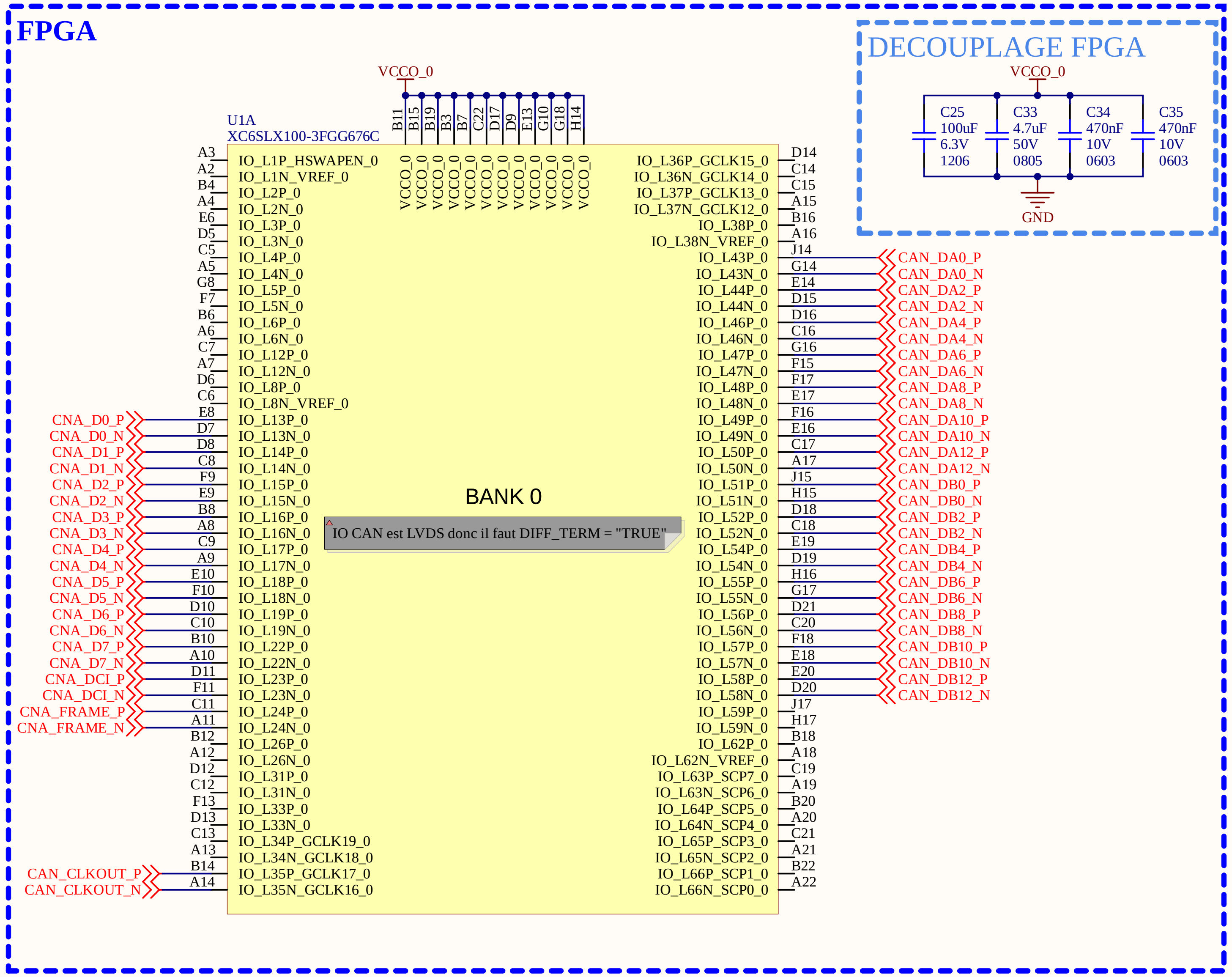


-Tous sur la BANK1
-ISENS = 1.65/(8x1,9A) = 0,108 Ohms

Sheet Name	MOTORS
Project Title	S7CAS-PLEIADES-CARTE-MERE
Global Project	*Pleiades*
Size	11x17
Group	01
Revision	1.00
Date	13/01/2026
Sheet	9 of 17
Filename	S7CAS-PLEIADES-CARTE-MERE_P01_MOTORS.SchDoc
Designers	Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard

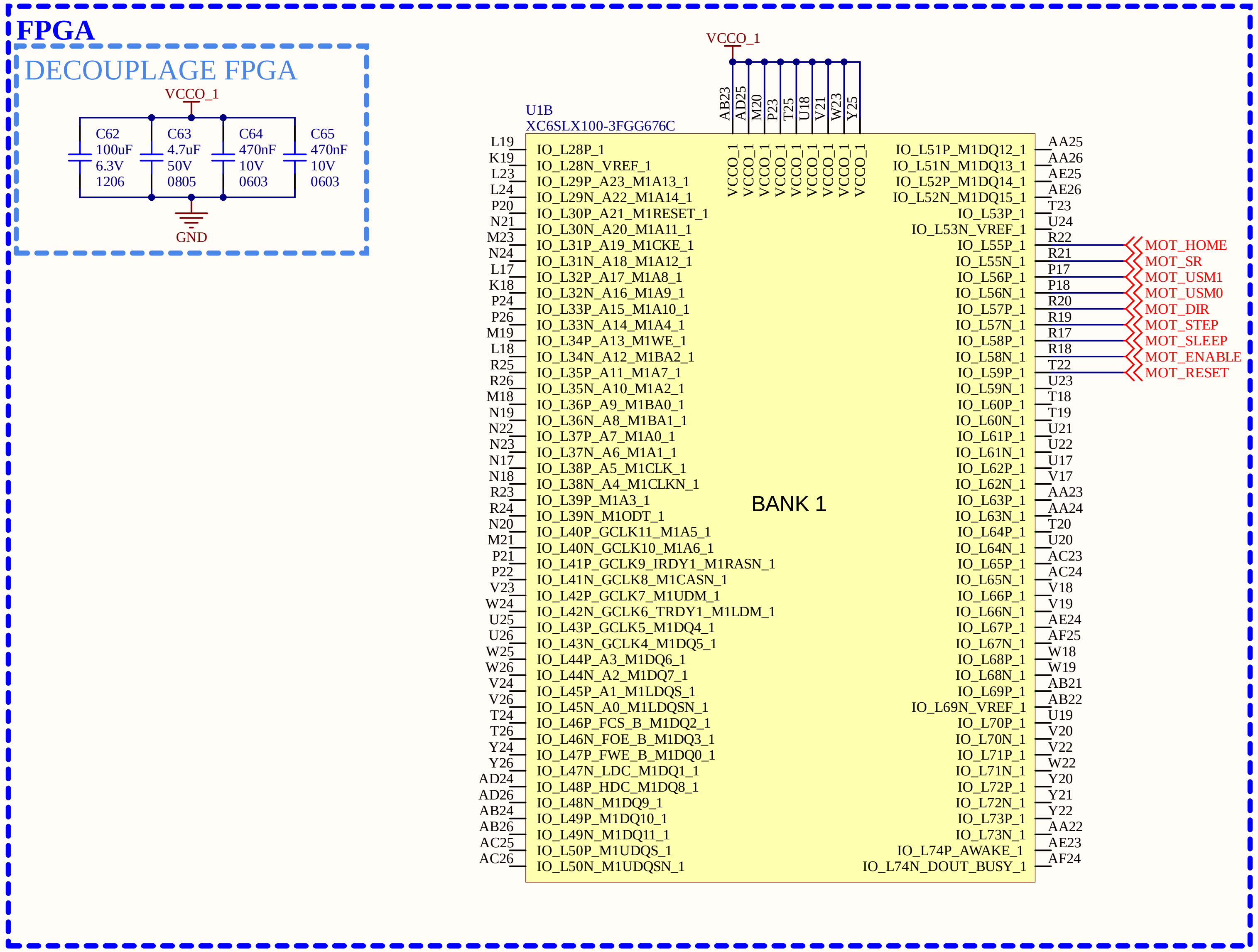


Sheet Name	IO		
Project Title	S7CAS-PLEIADES-CARTE-MERE		
Global Project	*Pleiades*		
Size: 11x17	Group: 01		Revision: 1.00
Date: 13/01/2026		Sheet 10 of 17	
Filename: S7CAS-PLEIADES-CARTE-MERE_P01_IO.SchDoc		Designers: Mathieu Desautels (DESM1210), Mathys Plante (PLAM1001), Jonathan Bouchard	



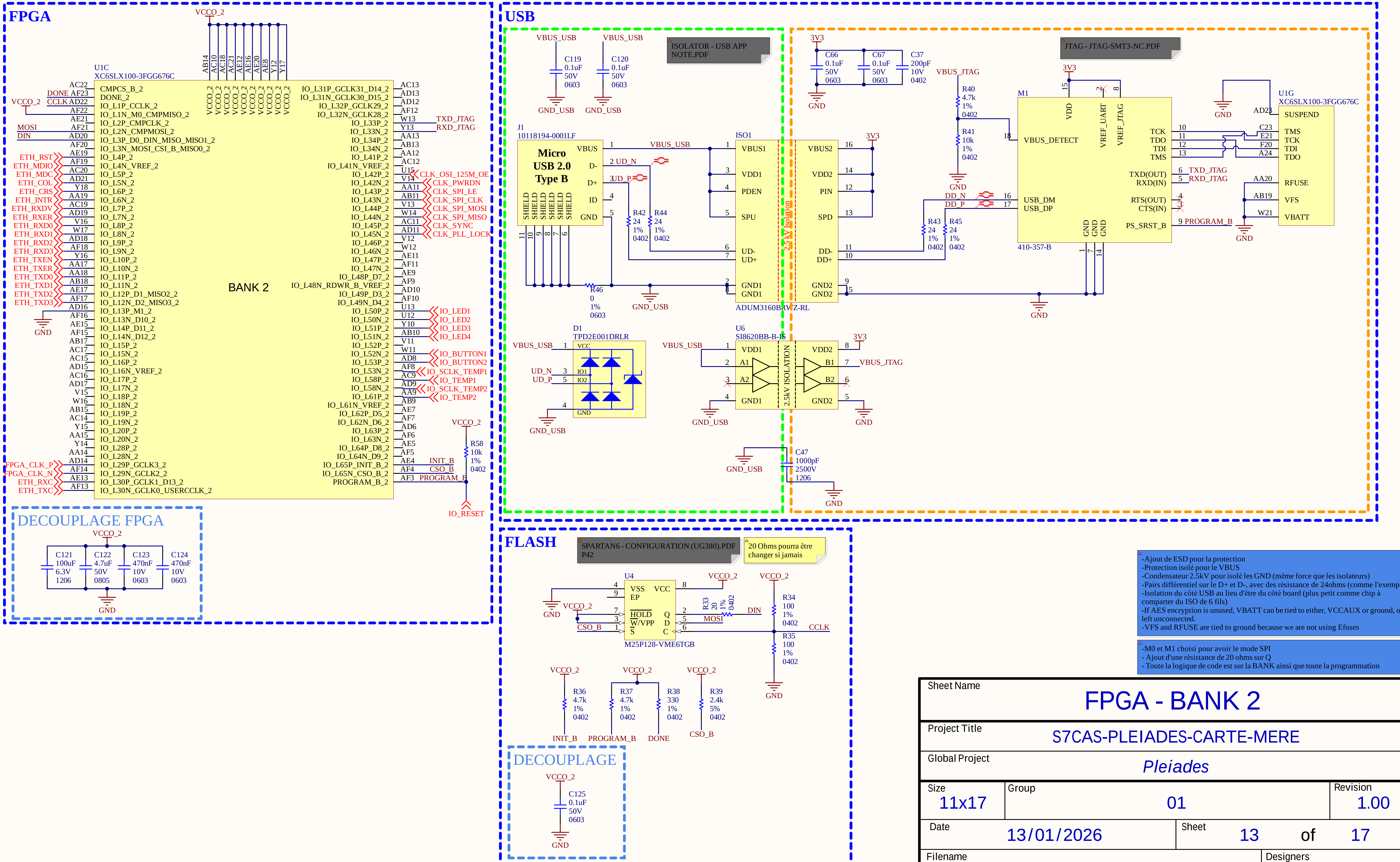
-LVDS seulement sur BANK0 et BANK2
-Toute les CAN et CNA sont mit sur cette BANK

Sheet Name	FPGA - BANK 0
Project Title	S7CAS-PLEIADES-CARTE-MERE
Global Project	*Pleiades*
Size	11x17
Group	01
Revision	1.00
Date	13/01/2026
Sheet	11 of 17
Filename	S7CAS-PLEIADES-CARTE-MERE_P01_FPGA-BANK0.SchDoc
Designers	Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard

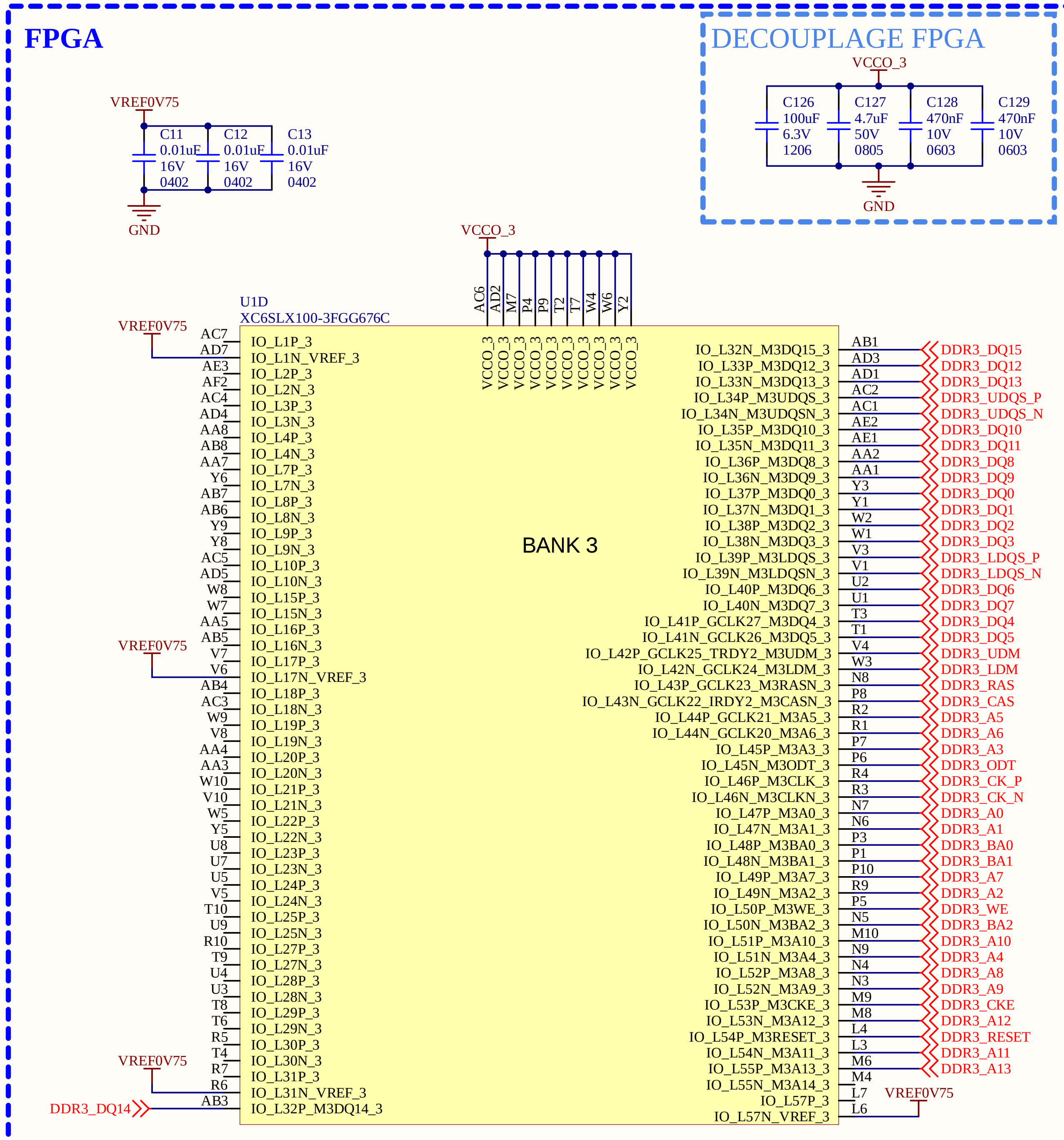


-Toute les controles de moteur sont sur cette BANK

Sheet Name	FPGA - BANK 1		
Project Title	S7CAS-PLEIADES-CARTE-MERE		
Global Project	*Pleiades*		
Size 11x17	Group 01		Revision 1.00
Date 13/01/2026		Sheet 12 of 17	
Filename S7CAS-PLEIADES-CARTE-MERE_P01_FPGA-BANK1.SchDoc		Designers Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard	

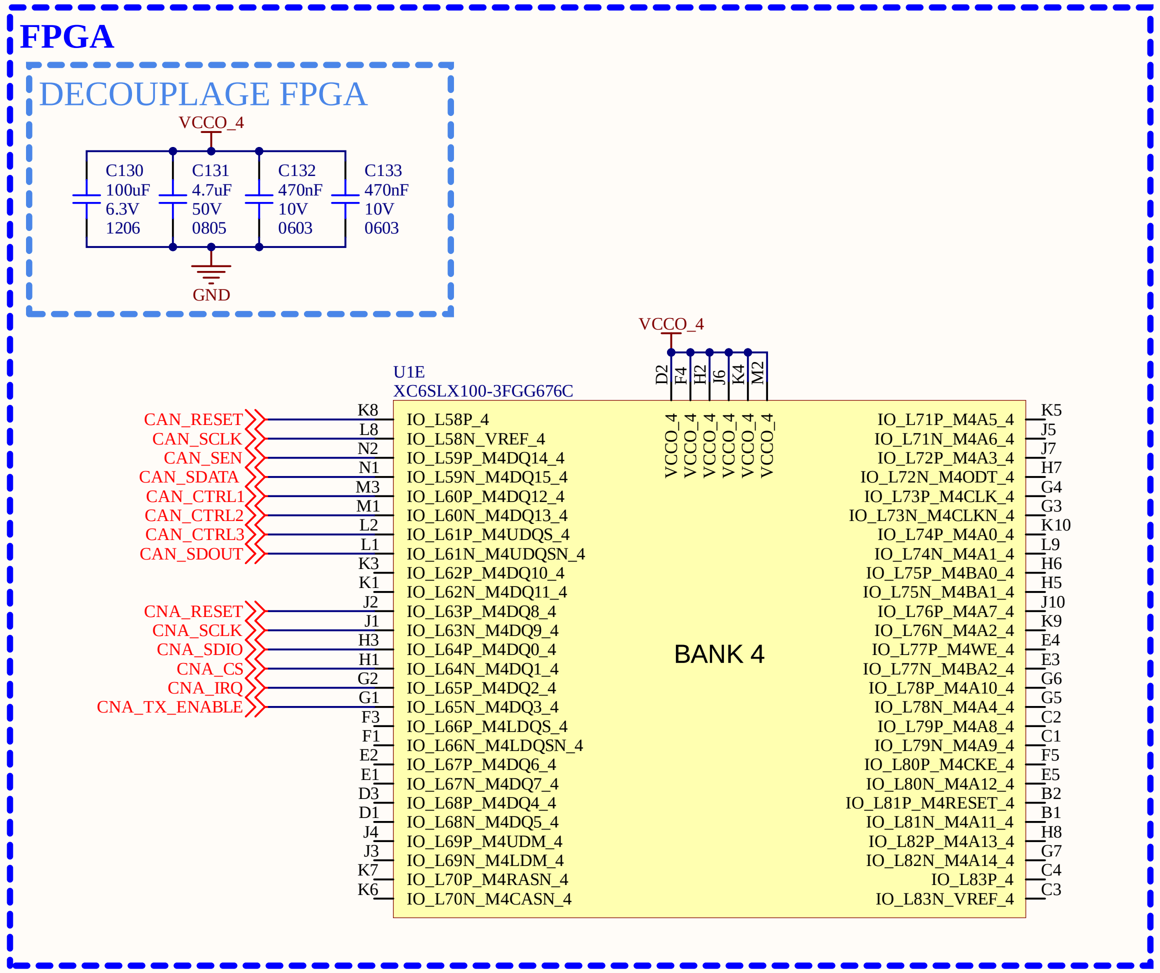


Sheet Name	FPGA - BANK 2
Project Title	S7CAS-PLEIADES-CARTE-MERE
Global Project	*Pleiades*
Size: 11x17	Group: 01 — Revision: 1.00
Date: 13/01/2026	Sheet 13 of 17
Filename: S7CAS-PLEIADES-CARTE-MERE_P01_FPGA-BANK2.SchDoc	Designers: Mathieu Desautels (DESM1210), Mathys Plante (PLAM1001), Jonathan Bouchard



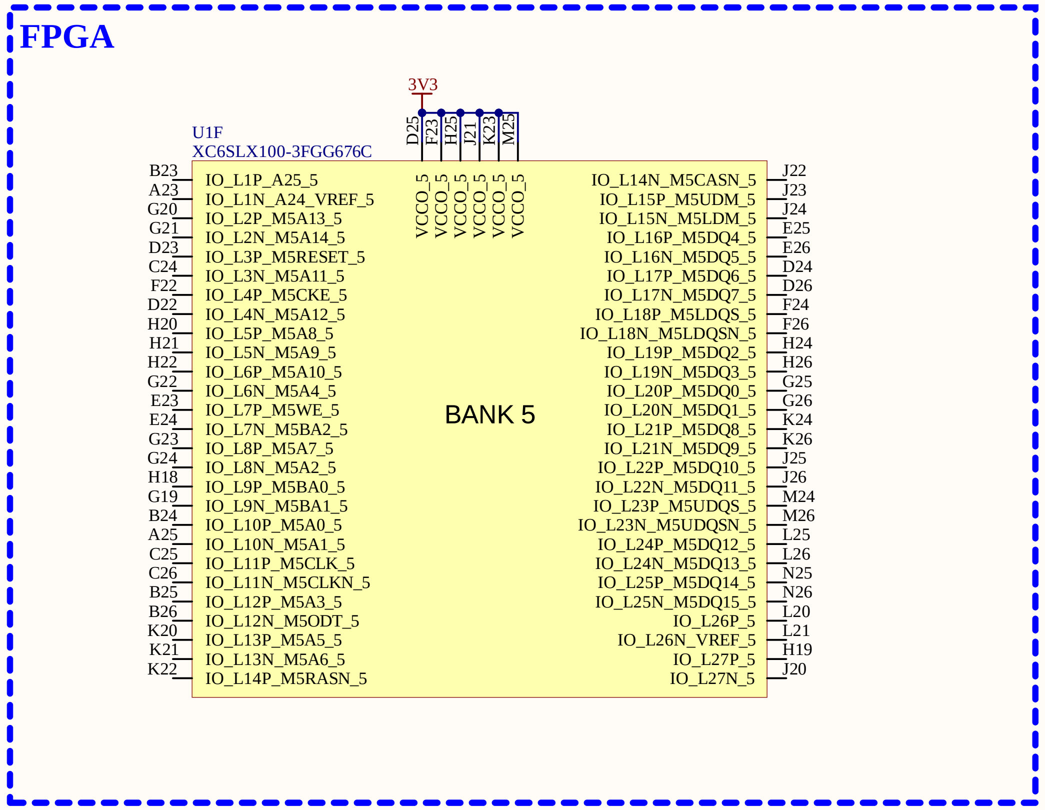
-Toute la DDR3 est sur cette BANK

Sheet Name	FPGA - BANK 3		
Project Title	S7CAS-PLEIADES-CARTE-MERE		
Global Project	*Pleiades*		
Size 11x17	Group 01		Revision 1.00
Date 13/01/2026		Sheet 14 of 17	
Filename S7CAS-PLEIADES-CARTE-MERE_P01_FPGA-BANK3.SchDoc		Designers Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard	



-Toute le controle du CNA et CAN sont sur cette BANK
-Leur tension n'est pas la même que celle de leur données

Sheet Name	FPGA - BANK 4
Project Title	S7CAS-PLEIADES-CARTE-MERE
Global Project	*Pleiades*
Size	11x17
Group	01
Revision	1.00
Date	13/01/2026
Sheet	15 of 17
Filename	S7CAS-PLEIADES-CARTE-MERE_P01_FPGA-BANK4.SchDoc
Designers	Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard



Sheet Name	FPGA - BANK 5
Project Title	S7CAS-PLEIADES-CARTE-MERE
Global Project	*Pleiades*
Size	11x17
Group	01
Revision	1.00
Date	13/01/2026
Sheet	16 of 17
Filename	S7CAS-PLEIADES-CARTE-MERE_P01_FPGA-BANK5.SchDoc
Designers	Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard

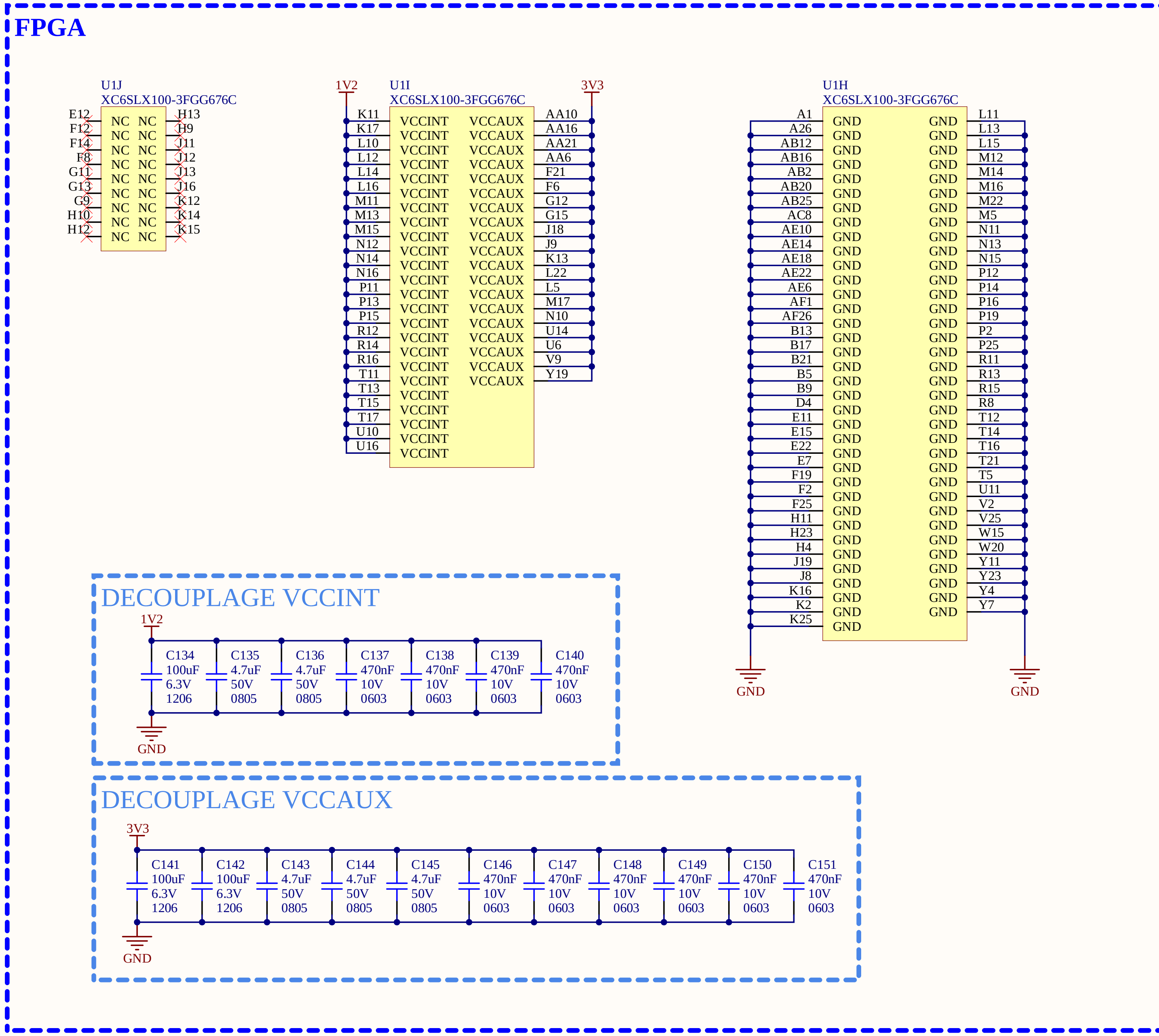


Table 2-1: **Required PCB Capacitor Quantities per Device[1][3] *(Continued)***

Package	Device (XC6S)	V_{CCINT} in µF			V_{CCAUX} in µF			V_{CCO} Bank 0 in µF			V_{CCO} Bank 1 in µF			V_{CCO} Bank 2 in µF			V_{CCO} Bank 3 in µF			V_{CCO} Bank 4 in µF			V_{CCO} Bank 5 in µF			Total [2]
		100	4.7	0.47	100	4.7	0.47	100	4.7	0.47	100	4.7	0.47	100	4.7	0.47	100	4.7	0.47	100	4.7	0.47	100	4.7	0.47	
FG(G)484	LX75	1	2	3	1	2	4	1	1	2	1	1	3	1	1	4	1	1	3							33
FG(G)484	LX75T	1	2	3	1	2	4	1	1	2	1	1	3	1	1	3	1	1	3							32
FG(G)484	LX100	1	2	4	1	2	4	1	1	2	1	1	3	1	1	3	1	1	3							33
FG(G)484	LX100T	1	2	4	1	2	4	1	1	2	1	1	3	1	1	3	1	1	3							33
FG(G)484	LX150	2	3	6	1	2	4	1	1	2	1	1	3	1	1	3	1	1	3							37
FG(G)484	LX150T	2	3	6	1	2	4	1	1	2	1	1	3	1	1	3	1	1	3							37
FG(G)676	LX45	1	1	2	1	2	5	1	1	3	1	1	3	1	1	3	1	1	4							33
FG(G)676	LX75	1	2	3	2	3	6	1	1	3	1	1	3	1	1	3	1	1	3	1	1	2	1	1	2	45
FG(G)676	LX75T	1	2	3	1	2	5	1	1	3	1	1	2	1	1	3	1	1	2	1	1	2	1	1	2	40
FG(G)676	LX100	1	2	4	2	3	6	1	1	3	1	1	3	1	1	3	1	1	3	1	1	2	1	1	2	46
FG(G)676	LX100T	1	2	4	1	2	5	1	1	3	1	1	2	1	1	3	1	1	2	1	1	2	1	1	2	41
FG(G)676	LX150	2	3	6	2	3	6	1	1	3	1	1	3	1	1	3	1	1	3	1	1	2	1	1	2	50
FG(G)676	LX150T	2	3	6	1	2	5	1	1	3	1	1	2	1	1	3	1	1	2	1	1	2	1	1	2	45
FG(G)900	LX100T	1	2	4	2	3	6	1	1	3	1	1	3	1	1	3	1	1	4	1	1	2	1	1	2	47
FG(G)900	LX150	2	3	6	2	4	7	1	1	5	1	1	3	1	1	5	1	1	4	1	1	2	1	1	2	57
FG(G)900	LX150T	2	3	6	2	3	7	1	1	4	1	1	3	1	1	4	1	1	4	1	1	2	1	1	2	54

Sheet Name	FPGA - ALIM
Project Title	S7CAS-PLEIADES-CARTE-MERE
Global Project	*Pleiades*
Size	11x17
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Sheet	17 of 17
Filename	S7CAS-PLEIADES-CARTE-MERE_P01_FPGA-ALIM.SchDoc
Designers	Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard